

AP Statistics FRQ Practice Exam 11

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APSTAT-FRQ11-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

FRQ Type: Formulate Questions and Collect Data

Prompt:

A university wants to estimate the proportion of its 6,000 commuter students who parked in an unauthorized campus area at least once during the previous month. Because the question is sensitive, the university takes a simple random sample of 300 commuter students and uses the following randomized-response procedure.

Each participating student privately rolls a fair six-sided die and does not reveal the result.

Die result 1 or 2: Answer “Yes.”

Die result 3 or 4: Answer “No.”

Die result 5 or 6: Answer the parking question truthfully.

Of the 300 selected students, 270 participate. Among the participants, 119 answer “Yes.”

(a) Identify the population, intended sample, respondent sample, and population parameter of interest.

(b) State a statistical question that the study is designed to answer. Describe how the university could select the original simple random sample from a complete commuter-student roster.

(c) Let p be the true proportion of commuter students who parked in an unauthorized area, and let q be the probability that a participant answers “Yes” under the procedure. Show that $q = 1/3 + p/3$. Explain how the procedure provides privacy.

(d) Use the observed responses to estimate p . Interpret the estimate in context.

(e) Explain why the randomized-response mechanism can reduce response bias but does not eliminate possible nonresponse bias. State a reasonable follow-up action.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Identifies the population as all 6,000 commuter students, the intended sample as the 300 selected students, and the respondent sample as the 270 participants.
- Point 2: Defines p as the true proportion of all 6,000 commuter students who parked in an unauthorized campus area at least once during the previous month.

(b) 2 points

- Point 1: States a question about the proportion or percentage of the university's commuter students who parked in an unauthorized area at least once during the previous month.
- Point 2: Describes a valid SRS method, such as assigning distinct labels 0001 through 6000 and using a random-number generator to select 300 distinct labels without replacement.

(c) 2 points

- Point 1: Derives $q = P(\text{forced Yes}) + P(\text{truthful branch and true Yes}) = 1/3 + (1/3)p$.
- Point 2: Explains that either response can result from the die mechanism or from truthful reporting, so the researcher cannot determine an individual student's parking behavior from the response alone.

(d) 2 points

- Point 1: Calculates $\hat{q} = 119/270 \approx 0.4407$ and $\hat{p} = 3\hat{q} - 1 \approx 0.3222$.
- Point 2: Interprets the estimate as approximately 32.2% of the university's commuter students having parked in an unauthorized area at least once during the previous month, subject to the study's limitations.

(e) 2 points

- Point 1: Explains that privacy may encourage more truthful responses, but the 30 nonparticipants could have a different unauthorized-parking rate, so randomized response does not remove nonresponse bias.
- Point 2: Proposes a reasonable action such as repeated confidential contact, varied contact times or modes, or a short nonresponse follow-up while preserving the randomized-response protocol.

Model Guidance:

The intended sample consists of everyone selected, whereas the respondent sample consists only of those who participated. The die creates known random noise: one-third of participants are forced to say Yes, one-third are forced to say No, and one-third answer truthfully. Thus, $q = 1/3 + p/3$, so the parking proportion is estimated by $3\hat{q} - 1$, not by the raw Yes proportion.

With $\hat{q} = 119/270$, the estimated parking proportion is about 0.322. The randomized mechanism protects individual privacy and can reduce deliberate misreporting. It cannot make the 270 respondents representative of the 30 nonrespondents, so confidential follow-up is still important.

Expected Points:

10

APSTAT-FRQ11-002

Section: FRQ

Type: frq

Primary Unit: Unit 5, Regression Analysis

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

An environmental team studies the relationship between streamside tree-canopy cover and midsummer water temperature at 10 monitoring sites in one watershed. For each site, x is the percentage of the stream segment covered by tree canopy and y is the mean afternoon water temperature, in degrees Celsius, during the monitoring week.

Site	A	B	C	D	E	F	G	H	I	J
x (%)	10	20	25	35	45	55	60	70	80	90
y (°C)	25.1	23.0	24.2	21.8	22.5	19.8	21.2	18.5	19.3	16.9

Least-squares output for predicting temperature from canopy cover is

$$\hat{y} = 25.756 - 0.09236x$$

$$r = -0.947$$

$$r^2 = 0.896$$

(a) Sketch a scatterplot with correctly labeled axes and an appropriate scale. Describe the form, direction, strength, and any unusual features of the relationship.

(b) For Site G, calculate the predicted temperature and residual. Interpret the sign of the residual in context.

(c) Interpret the slope and r^2 in context.

(d) According to the fitted line, determine the minimum whole-number canopy percentage needed for a predicted mean afternoon temperature of at most 19°C. Explain whether this use is interpolation or extrapolation.

(e) A watershed manager claims that increasing any site's canopy cover to the percentage found in part (d) will cause its mean afternoon temperature to be at most 19°C. Evaluate the claim using the data and study design.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Sketches a scatterplot with canopy cover on the horizontal axis, temperature on the vertical axis, suitable scales, and all 10 points plotted consistently with the table.
- Point 2: Describes a strong, negative, roughly linear relationship with some scatter and no clearly isolated unusual point.

(b) 2 points

- Point 1: Calculates $\hat{y} = 25.756 - 0.09236(60) \approx 20.214^\circ\text{C}$ and residual $= 21.2 - 20.214 \approx 0.986^\circ\text{C}$.

· Point 2: Interprets the positive residual to mean that Site G was about 0.99°C warmer than the model predicted for a site with 60% canopy cover.

(c) 2 points

· Point 1: Interprets the slope: each additional percentage point of canopy cover is associated with a predicted decrease of about 0.092°C in mean afternoon water temperature, on average.

· Point 2: Interprets $r^2 = 0.896$ as 89.6% of the variation in mean afternoon temperature among these sites being explained by the linear relationship with canopy-cover percentage.

(d) 2 points

· Point 1: Solves $25.756 - 0.09236x \leq 19$ to obtain $x \geq$ about 73.14%, and therefore selects 74% as the minimum whole-number percentage.

· Point 2: Identifies the calculation as interpolation because 74% lies within the observed canopy range of 10% to 90%.

(e) 2 points

· Point 1: Explains that the line predicts an average response and does not guarantee that every site at 74% canopy has a temperature at or below 19°C , as shown by nonzero residual variation.

· Point 2: Rejects the causal wording because canopy cover was observed rather than randomly assigned and identifies a plausible confounder, such as stream depth, flow, elevation, or channel width, that could affect temperature and be associated with canopy.

Model Guidance:

The plotted points should descend from upper left to lower right in a nearly linear pattern. At 60% canopy, the model predicts about 20.214°C ; the observed 21.2°C is higher, so the residual is positive. The slope describes an association per percentage point, and r^2 describes explained variation in the response—not the percentage of sites lying on the line.

Solving the fitted equation gives a threshold just above 73%, so the minimum whole-number model setting is 74%. That value is inside the data range. Still, predicted means are not guarantees for individual sites, and an observational relationship cannot establish that changing canopy alone causes the temperature difference.

Expected Points:

10

APSTAT-FRQ11-003

Section: FRQ

Type: frq

Primary Unit: Unit 3, Inference for Categorical Data: Proportions

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

A public-library system has 12,000 members whose accounts have been inactive for at least one year. The system takes a simple random sample of 600 such members and randomly assigns 300 to receive a personalized reactivation message and 300 to receive a standard message. A member is classified as reactivated if the member borrows at least one item during the next 30 days.

Message type	Reactivated	Not reactivated	Total
Personalized	84	216	300
Standard	63	237	300

At the $\alpha = 0.05$ significance level, test whether the personalized message produces a higher reactivation proportion than the standard message.

- Define the population proportions and state the null and alternative hypotheses.
- Identify the appropriate inference procedure and check its conditions.
- Calculate the sample proportions, pooled proportion, standard error, test statistic, and p-value.
- State a conclusion in context using $\alpha = 0.05$, including the appropriate scope of inference.
- Describe a Type I error in context.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Defines p_P and p_S as the true reactivation proportions for all 12,000 inactive members if sent the personalized and standard messages, respectively.
- Point 2: States $H_0: p_P - p_S = 0$ and $H_a: p_P - p_S > 0$.

(b) 3 points

- Point 1: Selects a one-sided two-proportion z test.
- Point 2: Cites the original SRS, random assignment to two separate groups, and the 10% condition because $600 \leq 0.10(12,000) = 1,200$.
- Point 3: Checks the pooled large-count condition using expected counts $300(0.245) = 73.5$ reactivated and $300(0.755) = 226.5$ not reactivated in each group, all at least 10.

(c) 3 points

- Point 1: Calculates $\hat{p}_P = 84/300 = 0.280$, $\hat{p}_S = 63/300 = 0.210$, observed difference 0.070, and pooled $\hat{p} = 147/600 = 0.245$.
- Point 2: Calculates $SE_0 = \sqrt{0.245(0.755)(1/300 + 1/300)} \approx 0.03512$ and $z = 0.070/0.03512 \approx 1.99$.
- Point 3: Calculates the one-sided p-value $P(Z \geq 1.99) \approx 0.0231$.

(d) 1 point

- Point 1: Because $0.0231 < 0.05$, rejects H_0 and concludes that there is convincing evidence that the personalized message causes a higher reactivation proportion than the standard message for the population of 12,000 inactive members, with causation supported by random assignment and generalization supported by the SRS.

(e) 1 point

- Point 1: Describes a Type I error as concluding that the personalized message produces a higher reactivation proportion when, in truth, the two message types have equal reactivation proportions for the

population.

Model Guidance:

This is a comparison of two independent treatment proportions. The null model uses one common proportion, so the standard error for the test must use the pooled estimate $147/600 = 0.245$. Random sampling supports population generalization, and random assignment supports a causal comparison of the two message types.

The observed advantage is 0.070, producing $z \approx 1.99$ and a one-sided p-value of about 0.0231. At $\alpha = 0.05$, the null hypothesis is rejected. A Type I error would mean asserting a real personalized-message advantage even though no population treatment difference exists.

Expected Points:

10

APSTAT-FRQ11-004

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: Unit 4, Inference for Quantitative Data: Means

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

A packaging machine fills more than 1,000 snack pouches each hour. When the process is operating as intended, individual fill masses are independent and Normally distributed with mean 250 grams and standard deviation 12 grams. The standard deviation remains 12 grams even if the process mean shifts.

Under the current monitoring rule, an independent random sample of 36 pouches is selected each hour. The machine is stopped for inspection if the sample mean fill mass is less than 246 grams.

(a) When the process mean is 250 grams, describe the sampling distribution of the sample mean and calculate the probability that the rule stops the machine. Interpret this probability as a false-alarm rate.

(b) Suppose the process mean has shifted to 245 grams. Calculate the probability that the current rule stops the machine and the probability that it misses the shift during a given hour.

(c) A supervisor says, "Because the false-alarm probability is only about 2.3%, about 97.7% of all stop alerts must indicate a real shift." Evaluate this statement.

(d) A proposed rule uses a sample of 64 pouches. Determine the cutoff c for stopping when $\bar{x} < c$ so that the false-alarm probability remains approximately 0.0228. Then calculate the probability that the proposed rule detects a shift to 245 grams.

(e) Compare the current and proposed rules. Recommend one rule and justify the recommendation using statistical evidence, operational cost, and the model assumptions.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: States that, when $\mu = 250$, $\bar{x}$ is Normal with mean 250 grams and standard deviation $12/\sqrt{36} = 2$ grams.
- Point 2: Calculates $P(\bar{x} < 246) = P(Z < -2) \approx 0.0228$ and interprets it as about a 2.28% chance of an unnecessary stop in an hour when the process mean is on target.

(b) 2 points

- Point 1: When $\mu = 245$, calculates $P(\bar{x} < 246) = P(Z < 0.50) \approx 0.6915$ as the hourly detection probability.
- Point 2: Calculates $1 - 0.6915 = 0.3085$ and interprets it as the probability that the current rule fails to stop during an hour when the true mean is 245 grams.

(c) 2 points

- Point 1: Rejects the statement because 0.0228 is $P(\text{alert} \mid \text{no shift})$, whereas the supervisor is making a claim about $P(\text{shift} \mid \text{alert})$, which reverses the conditioning.
- Point 2: Explains that the proportion of alerts corresponding to real shifts also depends on how frequently process shifts occur, information that is not provided.

(d) 2 points

- Point 1: Uses the new standard deviation $12/\sqrt{64} = 1.5$ and $z = -2$ to obtain $c = 250 - 2(1.5) = 247$ grams.
- Point 2: When $\mu = 245$, calculates $P(\bar{x} < 247) = P[Z < (247 - 245)/1.5] = P(Z < 1.33) \approx 0.9088$.

(e) 2 points

- Point 1: Gives an evidence-based comparison: both rules have about the same 2.28% false-alarm rate, but the proposed rule increases detection probability from about 0.6915 to 0.9088 and reduces the miss probability.
- Point 2: Gives a justified recommendation that weighs the proposed rule's improved detection against the cost of inspecting 64 rather than 36 pouches and notes reliance on random independent sampling, approximate Normal behavior, and a stable 12-gram standard deviation.

Model Guidance:

The sample mean has standard deviation $\sigma/\sqrt{n}$. For $n = 36$, the standard error is 2 grams, so a 246-gram cutoff is two standard errors below the on-target mean and gives a false-alarm probability of 0.0228. If the process shifts to 245 grams, the same cutoff is 0.5 standard error above the new mean, giving detection probability 0.6915 and miss probability 0.3085.

The supervisor reverses a conditional probability: a low false-alarm rate does not determine the percentage of alerts that are true without a process-shift base rate. With $n = 64$, a cutoff of 247 grams preserves the $z = -2$ false-alarm boundary while raising detection probability for $\mu = 245$ to about 0.9088. The larger-sample rule has better statistical performance for that shift, but a recommendation should also account for inspection cost and whether the Normal, independence, and constant-variability assumptions are credible.

Expected Points:

10

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ11

- APSTAT-FRQ11-001: Unit 1 primary; Unit 2 secondary; SRS of 300 of 6,000 commuter students with 270 respondents using a six-sided-die randomized-response design for unauthorized parking, with forced Yes/forced No/truthful branches; derive $q = 1/3 + p/3$, estimate $p = 3(119/270) - 1 \approx 0.322$, explain privacy, and distinguish response-bias reduction from unresolved nonresponse bias; targets using the raw Yes rate as the sensitive-behavior estimate and assuming randomized response repairs nonresponse.
- APSTAT-FRQ11-002: Unit 5 primary; Unit 1 secondary; raw data for 10 watershed sites relating canopy-cover percentage to mean afternoon stream temperature, with $\hat{y} = 25.756 - 0.09236x$, $r = -0.947$, and $r^2 = 0.896$; sketch and describe the scatterplot, compute Site G prediction and residual, interpret slope and r^2 , solve a 19°C prediction threshold at 74% canopy, and reject guaranteed or causal interpretation of an observational model; targets treating explained variation as a percentage of points and converting association into certainty or causation.
- APSTAT-FRQ11-003: Unit 3 primary; Unit 1 secondary; SRS of 600 of 12,000 inactive library members randomly assigned 300 each to personalized and standard reactivation messages, with 84 versus 63 reactivations; conduct a pooled one-sided two-proportion z test with difference 0.070, pooled $\hat{p} = 0.245$, $z \approx 1.99$, $p \approx 0.0231$ at $\alpha = 0.05$, causal and population scope, and Type I error; targets using an unpooled test standard error and confusing statistical significance with an unsupported scope.
- APSTAT-FRQ11-004: Unit 2 primary; Unit 4 secondary; snack-pouch process with individual $\text{Normal}(\mu, 12)$ fills and hourly sample-mean stopping rule; for $n = 36$ and cutoff 246, calculate 0.0228 false-alarm probability at $\mu = 250$ and 0.6915 detection at $\mu = 245$, expose reversed conditional reasoning about alerts, then design $n = 64$ cutoff 247 preserving false alarms while raising detection to about 0.9088; targets confusing $P(\text{alert} \mid \text{no shift})$ with $P(\text{no shift} \mid \text{alert})$ and assuming larger samples improve performance without a specified cutoff rule.